Lab: Mapping the Markov Blanket of Your Invention

Objective

By the end of this lab, you will be able to identify and diagram the system boundaries of any invention — distinguishing internal states, external states, sensory boundaries, and active boundaries. You will apply this analysis to your own invention idea, regardless of your background or domain.

Materials and Prerequisites

- Paper and pen (or a digital drawing tool)
- Your own invention idea (any domain — a kitchen tool, a software app, a teaching method, an art installation, a community program, anything)
- If you do not yet have an invention idea, you will develop one in Part 1
- No programming or engineering background required

Part 1: Choosing and Describing Your Invention (10 minutes)

If you already have an invention idea, describe it briefly. If not, think about a frustration, inefficiency, or unmet need in your daily life or work. Your invention does not need to be high-tech — a new way to organize a pantry, a method for teaching children fractions, or a tool for applying paint evenly are all valid inventions.

Describe your invention idea in 2-3 sentences:

Write your response here

What problem does it solve? Who experiences this problem?

Write your response here

What domain does your invention belong to? (e.g., cooking, education, healthcare, art, software, construction, music, agriculture, etc.)

Write your response here

Part 2: Identifying System Components (15 minutes)

Now we will decompose your invention into its system components using the Active Inference framework.

Step 2a: Internal States

Internal states are everything "inside" your invention — the parts, mechanisms, logic, or processes that make it work. These are the states that are hidden from the environment by the Markov blanket.

List 5-8 internal states of your invention:

#	Internal State	What it does
1		

Write your response here

|

Write your response here

| | 2 |

Write your response here

|

Write your response here

| | 3 |

Write your response here

|

Write your response here

| | 4 |

Write your response here

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Write your response here

| | 5 |

Write your response here

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Write your response here

| | 6 |

Write your response here

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Write your response here

| | 7 |

Write your response here

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Write your response here

| | 8 |

Write your response here

|

Write your response here

|

Step 2b: External States

External states are everything in the environment that affects or is affected by your invention, but is not part of the invention itself.

List 5-8 external states relevant to your invention:

#	External State	How it relates to your invention
1		

Write your response here

|

Write your response here

| | 2 |

Write your response here

|

Write your response here

| | 3 |

Write your response here

|

Write your response here

| | 4 |

Write your response here

|

Write your response here

| | 5 |

Write your response here

|

Write your response here

| | 6 |

Write your response here

|

Write your response here

| | 7 |

Write your response here

|

Write your response here

| | 8 |

Write your response here

|

Write your response here

|

Step 2c: Sensory Boundary

Sensory states are the parts of the Markov blanket through which your invention receives information from the environment.

List 3-5 sensory states (how your invention "senses" the world):

#	Sensory State	What information it captures
1		

Write your response here

|

Write your response here

| | 2 |

Write your response here

|

Write your response here

| | 3 |

Write your response here

|

Write your response here

| | 4 |

Write your response here

|

Write your response here

| | 5 |

Write your response here

|

Write your response here

|

Step 2d: Active Boundary

Active states are the parts of the Markov blanket through which your invention changes the environment.

List 3-5 active states (how your invention "acts on" the world):

#	Active State	What change it produces
1		

Write your response here

|

Write your response here

| | 2 |

Write your response here

|

Write your response here

| | 3 |

Write your response here

|

Write your response here

| | 4 |

Write your response here

|

Write your response here

| | 5 |

Write your response here

|

Write your response here

|

Part 3: Drawing the System Diagram (15 minutes)

On a blank sheet of paper (or digital canvas), draw the following diagram of your invention:

1. Draw a large circle or rectangle in the center — this represents your invention's Markov blanket.
2. Inside the shape, write or draw your internal states.
3. Outside the shape, write or draw your external states.
4. On the left side of the boundary, draw arrows pointing INWARD for each sensory state. Label them.
5. On the right side of the boundary, draw arrows pointing OUTWARD for each active state. Label them.
6. Draw a curved arrow from the active states, through the external environment, back to the sensory states — this is the perception-action loop.

After drawing your diagram, answer these questions:

Does the perception-action loop close? That is, do the actions of your invention eventually lead to changes that are sensed back by the invention (directly or through the user)?

Write your response here

Are there any external states that your invention should sense but currently does not? What information is missing?

Write your response here

Are there any actions your invention should take but currently cannot? What capabilities are missing?

Write your response here

Part 4: Analyzing Nested Systems (10 minutes)

Most inventions contain subsystems. Identify at least two subsystems within your invention, and analyze them as systems in their own right.

Subsystem 1:

Property	Description
Name	

Write your response here

| | Internal states |

Write your response here

| | Sensory boundary |

Write your response here

| | Active boundary |

Write your response here

| | What it is conditionally independent of |

Write your response here

|

Subsystem 2:

Property	Description
Name	

Write your response here

| | Internal states |

Write your response here

| | Sensory boundary |

Write your response here

| | Active boundary |

Write your response here

| | What it is conditionally independent of |

Write your response here

How do these two subsystems interact? Do they share any boundary states?

Write your response here

Part 5: Stress-Testing Your Boundaries (10 minutes)

Now consider what happens when your invention encounters unexpected conditions. This tests whether your system boundaries are drawn correctly.

Scenario 1: An environmental change your invention was NOT designed for.

Describe a plausible unexpected condition:

Write your response here

How would this condition affect your invention's internal states?

Write your response here

Does your current sensory boundary detect this condition? If not, what sensor or input would you add?

Write your response here

Scenario 2: A user who uses your invention differently than intended.

Describe an unintended use:

Write your response here

Would your current system boundary accommodate this use, or would you need to redraw the boundary?

Write your response here

Scenario 3: Scaling up.

If you needed to produce 1000 of your inventions (or serve 1000 users), which system boundaries would change?

Write your response here

Discussion and Debrief

Reflect on the following questions individually or with your group:

1. **Boundary placement**: Where did you find it hardest to decide what was "inside" vs. "outside" your invention? What does this ambiguity tell you about your design?

2. **Sensory gaps**: Most early-stage inventions have significant gaps in their sensory boundary — they do not sense important environmental variables. What were the biggest sensory gaps in your design, and how could you close them?

3. **The perception-action loop**: Is the loop in your invention closed tightly (fast feedback) or loosely (delayed feedback)? How does the loop speed affect your invention's ability to function well?

4. **Nested complexity**: How many levels of nesting does your invention require? Is there a way to simplify the nesting while preserving functionality?

5. **Boundary lessons**: How might the exercise of formally mapping system boundaries change how you approach your next design decision?

Write your response here